

ACME VCF 9 Upgrade - SOW v1

Statement of Work

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Executive Summary

ACME is undertaking a strategic brownfield transformation of its existing VMware estate — comprising 9 standalone vCenter instances across 3 sites, approximately 64–74 ESXi hosts, and 3,500–4,000 virtual machines — onto a single VMware Cloud Foundation (VCF) 9 instance. The current environment consists of isolated vCenter/ESXi islands running vCenter 8.x, with mixed vxLAN (NSX-T confirmed) and VLAN networking, and external SAN storage. VCF 9 licensing has been purchased. The engagement will deliver a greenfield VCF 9 build comprising one management domain and separate workload domains for Production, UAT, and DR, with DR and HA capabilities required across all 3 sites. Existing workloads will be migrated into the new platform rather than upgraded in place, with all activity subject to the strict change window constraints typical of a financial institution.

This engagement delivers a consolidated, software-defined infrastructure foundation that eliminates operational silos, introduces consistent lifecycle management via SDDC Manager, and positions ACME for long-term platform agility. Key workstreams include hardware HCL validation, NSX-T to VCF 9 network design, SAN supplemental storage integration, management domain consolidation from 3 vCenters into a single SDDC Manager-governed construct, structured VM migration sequenced within bank change windows, multi-site DR and HA design, and a dedicated knowledge transfer programme to address the team's current zero-experience baseline with SDDC Manager. Success is defined as all applications functioning as normal or better following migration to the VCF 9 platform.

Objectives

The following measurable objectives define the success criteria for the ACME VCF 9 Upgrade engagement. Each objective is directly traceable to confirmed requirements, facts, and assumptions.

Platform Consolidation

- Deploy a single, production-ready VMware Cloud Foundation 9 instance comprising one management domain and separate workload domains for Production, UAT, and DR, replacing 9 existing standalone vCenter islands.
- Consolidate 3 existing management vCenter instances into a single SDDC Manager-managed VCF 9 management domain across all 3 sites.
- Commission a minimum of 3 workload domains (Production, UAT, DR) within the VCF 9 instance, aligned to the existing vCenter island topology.

Infrastructure & Hardware Readiness

- Complete a hardware HCL validation assessment for all ~64–74 hosts against VCF 9 / ESXi 9 requirements (CPU generation, NIC, HBA, firmware) prior to deployment, with a documented remediation plan for any non-compliant hosts.
- Validate or remediate all hosts to meet VCF 9 management domain and workload domain minimum hardware requirements before greenfield build commences.
- Confirm SAN storage protocol (FC or iSCSI) and design SAN integration as supplemental storage within VCF 9 workload domains, with all design decisions documented and agreed prior to deployment.

Networking

- Produce a complete network mapping of the existing mixed vxLAN (NSX-T) and VLAN environment across all 3 sites, and deliver a validated NSX 4.x overlay and underlay design for VCF 9.
- Execute NSX-T to NSX 4.x migration (as bundled in VCF 9) with zero unplanned production network outages, validated by post-migration connectivity testing.

Workload Migration

- Migrate all 3,500–4,000 VMs from existing standalone vCenter islands into the appropriate VCF 9 workload domains, sequenced within approved bank change windows.
- Achieve application-level validation confirming all migrated applications are functioning as normal or better post-migration, as defined in the agreed acceptance criteria.

DR & HA

- Design and implement DR and HA capabilities across all 3 sites within the VCF 9 platform, with documented RPO/RTO targets agreed with the customer prior to deployment.
- Conduct at least one documented DR failover test per site pairing, with results reviewed and signed off by the customer.

Operational Readiness

- Deliver structured SDDC Manager knowledge transfer and admin training to the customer operations team, covering LCM, workload domain commissioning, and day-2 operations, prior to project handover.
- Produce a complete set of as-built documentation and operational runbooks for the VCF 9 environment.
- Achieve formal customer sign-off at the conclusion of UAT, confirming the environment meets all agreed acceptance criteria before handover.

High-Level Scope of Work

This engagement delivers a greenfield VMware Cloud Foundation 9 platform across three on-premises sites, consolidating nine existing standalone vCenter islands into a single VCF 9 instance with a unified management domain and discrete workload domains for Production, UAT, and DR. The scope covers infrastructure assessment and design, VCF 9 platform build, NSX-T overlay migration, SAN storage integration, workload migration, DR/HA configuration, and structured knowledge transfer.

Phase 1 — Assessment & Design

This phase establishes the authoritative baseline of the existing environment and produces all design artefacts required to begin deployment. No infrastructure changes are made during this phase.

Activities

- Conduct a full hardware inventory across all ~64–74 hosts (4 management + ~60–70 workload) and validate each against the VCF 9 / ESXi 9 Hardware Compatibility List (HCL), with specific focus on CPU generation (Skylake or later required), NIC models, and HBA types
- Identify any pre-Skylake hosts that will require hardware refresh before VCF 9 deployment can proceed; produce a hardware gap report with remediation options
- Validate NIC and HBA firmware and driver versions against VCF 9 HCL requirements
- Inventory all 9 existing vCenter instances (3 management, 2 production, 2 UAT, 2 DR), document cluster topology, resource pools, distributed switches, and host profiles
- Perform VM inventory across the estate (confirmed 3,500–4,000 VMs), capturing OS type, storage profile, network segment, and application tier to inform migration wave planning
- Audit existing NSX-T configuration: transport zones, segments, logical routers, distributed firewall rule sets, and gateway policies across the mixed vxLAN/VLAN environment
- Map all existing VLAN and vxLAN segments to a target NSX 4.x (VCF 9 bundled) overlay design
- Assess SAN storage environment: document protocol (FC or iSCSI — TBC), array model, LUN layout, multipathing policy, and connectivity per site; validate against VCF 9 supplemental storage requirements
- Assess vSAN readiness of management domain hosts (drive types, cache/capacity tier eligibility) even where SAN is retained as primary workload storage
- Conduct a VCF 9 component version review to confirm SDDC Manager, NSX, vSAN, and ESXi 9 build availability and stability for production deployment
- Design the VCF 9 management domain: single SDDC Manager instance, consolidation of 3 site-level management vCenters into one VCF-managed construct, and multi-site topology (one cluster per site or stretched design — to be confirmed)
- Design workload domains: Production, UAT, and DR — each mapped to the appropriate existing vCenter pairs, with host-to-domain assignment, cluster sizing, and resource allocation defined
- Design DR and HA architecture across all 3 sites: define RPO/RTO targets, site roles (primary/secondary/tertiary), vSphere HA policy per domain, and DR tooling selection (e.g. VMware Live Site Recovery or equivalent)
- Design NSX 4.x overlay for VCF 9: transport node profiles, segment design, Tier-0/Tier-1 gateway topology, distributed firewall policy migration approach, and BGP/routing integration
- Design SAN integration as supplemental storage within VCF 9 workload domains; document any VCF 9 feature limitations imposed by non-vSAN primary storage
- Produce migration wave plan: group 3,500–4,000 VMs into sequenced waves by workload domain, application tier, and criticality; align waves to bank change windows
- Produce the High-Level Design (HLD) and Low-Level Design (LLD) documents for review and sign-off

- Conduct design review workshop with customer technical and architecture stakeholders

Deliverables

- Hardware HCL Assessment Report (per-host CPU, NIC, HBA validation; refresh requirements identified)
 - VM & Workload Inventory Report (3,500–4,000 VMs, segmented by domain and migration wave)
 - NSX-T Current-State Audit Report (segment map, DFW rule inventory, routing topology)
 - SAN Storage Assessment Report (protocol, array, LUN layout, VCF 9 supplemental storage compatibility)
 - VCF 9 High-Level Design (HLD) — management domain, workload domains, multi-site topology, DR/HA architecture
 - VCF 9 Low-Level Design (LLD) — SDDC Manager configuration, NSX 4.x design, storage design, cluster specifications
 - Migration Wave Plan — VM groupings, sequencing, tooling selection, and change window alignment
 - Design Sign-Off Record
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Phase 2 — VCF 9 Platform Build (Management Domain)

This phase deploys the VCF 9 management domain across the three sites, establishing the SDDC Manager control plane before any workload domains or migrations begin.

Activities

- Prepare management domain hosts: validate HCL compliance, apply required firmware and driver updates, configure physical networking (uplinks, MTU, LACP where applicable)
- Deploy VCF 9 Cloud Builder and execute management domain bring-up using the validated LLD
- Configure SDDC Manager: licensing, depot connectivity, certificate management, and identity source integration (Active Directory/LDAP)
- Deploy and configure NSX 4.x management plane components (NSX Manager cluster) as part of management domain bring-up
- Configure management domain vCenter: datacenter/cluster objects, DRS/HA policy, and host profiles aligned to LLD
- Integrate management domain with SAN storage (supplemental storage configuration per VCF 9 design)
- Validate management domain health via SDDC Manager: all components green, LCM depot reachable, certificate chain valid
- Configure multi-site management domain topology: establish site awareness, vCenter cluster-per-site or stretched configuration per agreed design
- Implement management domain DR/HA configuration: vSphere HA policy, admission control, and site-level failover behaviour
- Conduct management domain functional testing: vCenter access, SDDC Manager operations, NSX Manager availability, storage connectivity

Deliverables

- Deployed and validated VCF 9 Management Domain (SDDC Manager, vCenter, NSX Manager)

- Management Domain Build Runbook (step-by-step record of deployment actions and validation checks)
 - Management Domain Test Evidence Report
 - SDDC Manager Configuration Baseline Document
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Phase 3 — Workload Domain Build (Production, UAT, DR)

This phase commissions the three workload domains within SDDC Manager and configures NSX overlay, storage, and DR/HA capabilities ready to receive migrated workloads.

Activities

- Commission workload domain hosts: HCL validation, firmware/driver alignment, physical network preparation
- Use SDDC Manager to commission Production workload domain: host onboarding, cluster creation, vCenter deployment, NSX transport node configuration
- Repeat workload domain commissioning for UAT and DR domains
- Configure NSX 4.x overlay per workload domain: transport zones, segments, Tier-0/Tier-1 gateways, BGP peering, and distributed firewall baseline policy
- Migrate existing NSX-T segment definitions and DFW rule sets from standalone NSX-T instances into the VCF 9 NSX 4.x fabric, validating policy equivalence before cutover
- Integrate SAN storage with each workload domain as supplemental storage: datastore presentation, multipathing policy, and storage policy definition in vCenter
- Configure vSphere HA per workload domain cluster: admission control policy, heartbeat datastores, and isolation response aligned to RPO/RTO targets
- Configure cross-site DR capabilities: define protection groups, recovery plans, and failover runbooks for Production and UAT domains; configure DR workload domain as recovery target
- Validate inter-site network connectivity and latency for DR replication and HA heartbeat paths
- Conduct workload domain functional testing: VM provisioning, storage I/O, network overlay connectivity, DFW enforcement, and DR failover test (non-production)

Deliverables

- Deployed and validated Production, UAT, and DR Workload Domains
 - NSX 4.x Overlay Configuration Record (segments, gateways, DFW policy baseline)
 - SAN Supplemental Storage Configuration Record (per workload domain)
 - DR/HA Configuration Document (protection groups, recovery plans, RPO/RTO mapping)
 - Workload Domain Build Runbook
 - Workload Domain Test Evidence Report
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Phase 4 — Workload Migration

This phase migrates 3,500–4,000 VMs from the nine standalone vCenter islands into the appropriate VCF 9 workload domains, executed in sequenced waves aligned to bank change windows.

Activities

- Finalise and communicate migration wave schedule with customer change management team; obtain change advisory board (CAB) approval for each wave
- Configure migration tooling per wave: vSphere vMotion (live, same-version), VMware HCX (where cross-version or cross-network migration is required), or cold migration for non-critical workloads — tooling selection per VM group as defined in the wave plan
- Execute Wave 1 (non-production / UAT VMs): migrate UAT workload VMs from standalone UAT vCenters into the VCF 9 UAT workload domain; validate application function post-migration
- Execute Wave 2 (Production VMs — Tier 2/3): migrate lower-criticality production workloads into the VCF 9 Production workload domain within approved change windows
- Execute Wave 3 (Production VMs — Tier 1 / business-critical): migrate highest-criticality workloads with dedicated rollback plans and application owner sign-off per VM group
- Execute DR domain population: migrate or replicate DR-designated VMs into the DR workload domain; validate replication and recovery plan integrity post-migration
- Perform post-wave network validation: confirm segment connectivity, DFW policy enforcement, and routing for each migrated VM group before proceeding to next wave
- Decommission source vCenter clusters progressively as waves complete and VMs are confirmed stable in VCF 9
- Maintain a live migration tracker (VM name, source vCenter, target domain, wave, status, sign-off) updated after each change window

Migration Approach Options

Option	Description	Advantages	Disadvantages	Relative Effort
A — Greenfield Build + vMotion/HCX Migration (Recommended)	Build VCF 9 alongside existing estate; migrate VMs using live vMotion or HCX in waves aligned to change windows	Lowest risk; rollback to source is available throughout; no production downtime required for most workloads; aligns with bank change window constraints	Requires parallel infrastructure capacity during transition; longer elapsed timeline	Medium-High
B — Greenfield Build + Cold Migration	Build VCF 9 alongside existing estate; power off VMs and cold-migrate storage during change windows	Simpler tooling; no HCX licensing required	Downtime required per VM; unsuitable for Tier-1 workloads; change window pressure increases	Medium

Option	Description	Advantages	Disadvantages	Relative Effort
C — In-Place Cluster Expansion then Migrate	Expand existing clusters with VCF 9-compliant hosts, then migrate VMs within the same vCenter before cutting over to SDDC Manager	Avoids parallel infrastructure cost	Highest complexity; risk of management plane disruption; not aligned to VCF 9 greenfield bring-up model; not recommended for brownfield consolidation of this scale	High

Recommended Approach: Option A. Building VCF 9 as a parallel greenfield instance and migrating workloads in sequenced waves using vMotion (same-version, same-network) and HCX (cross-network or cross-version) provides the lowest risk profile, preserves rollback capability throughout the migration, and is most compatible with bank change window constraints. Wave sequencing (UAT → Tier 2/3 Production → Tier 1 Production → DR) ensures lessons learned in lower-risk waves inform the approach for critical workloads.

Deliverables

- Approved Migration Wave Schedule (CAB-aligned)
- Migration Tooling Configuration Record (HCX deployment, vMotion network configuration)
- Per-Wave Migration Completion Reports (VM count, status, issues, sign-off)
- Post-Migration Network Validation Report
- Source vCenter Decommission Log
- Live Migration Tracker (final state)

Phase 5 — DR Validation & HA Testing

This phase formally validates the DR and HA capabilities across all three sites against agreed RPO/RTO targets.

Activities

- Execute planned DR failover test for Production workload domain: fail over protection group to DR site, validate application availability, and fail back; document actual RTO achieved
- Execute HA failover simulation: isolate a host within a workload domain cluster, validate vSphere HA restart behaviour and timing
- Test cross-site HA heartbeat and admission control behaviour under simulated site-link degradation
- Validate DR workload domain recovery plan completeness: confirm all Tier-1 VMs are included in protection groups with correct boot order and network mapping
- Document RPO/RTO results against targets defined in Phase 1 design; raise remediation actions for any gaps
- Conduct site-level failover tabletop exercise with customer operations team

Deliverables

- DR Failover Test Report (planned failover, failback, RTO measurement)
 - HA Validation Test Report
 - RPO/RTO Compliance Summary
 - DR Runbook (customer-facing, operational)
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Phase 6 — UAT, Knowledge Transfer & Hypercare

This phase obtains formal customer acceptance of the VCF 9 platform and addresses the identified SDDC Manager operational readiness gap through structured knowledge transfer and an extended hypercare period.

Activities

- Support customer UAT execution: provide technical assistance for application-level validation across migrated workloads; success criterion is applications functioning as normal or better post-migration
- Conduct structured SDDC Manager administration training covering: LCM (lifecycle management) operations, workload domain commissioning and decommissioning, host onboarding/offboarding, certificate management, and depot/patch management
- Conduct NSX 4.x day-2 operations training: segment management, DFW rule administration, gateway policy changes, and troubleshooting
- Conduct VCF 9 day-2 operations workshop: vSphere HA/DRS tuning, storage policy management, SAN supplemental storage operations, and SDDC Manager alerting
- Deliver all project documentation to customer: HLD, LLD, build runbooks, migration records, DR runbooks, and test evidence
- Provide extended hypercare support (duration to be agreed — recommended minimum 4 weeks post-handover) covering: break-fix assistance for VCF 9 platform issues, guidance on first LCM patch cycle, and escalation support for operational queries
- Conduct formal project close-out and lessons-learned session
- Obtain customer sign-off on project completion

Deliverables

- UAT Sign-Off Record (application validation, customer-signed)
- SDDC Manager Administration Training Materials and Attendance Record
- NSX 4.x Day-2 Operations Training Materials
- VCF 9 Day-2 Operations Runbook (customer-facing)
- Complete Documentation Pack (HLD, LLD, all build and test records)
- Hypercare Support Log
- Project Completion and Lessons Learned Report
- Customer Project Sign-Off Certificate

Migration / Implementation Options

Three viable approaches exist for migrating ACME's brownfield vCenter estate into VCF 9. Each is evaluated below against the confirmed constraints: 9 existing vCenter instances (3 management, 2 production, 2 UAT, 2 DR), ~64-74 hosts across 3 sites, ~3,500-4,000 VMs, SAN-based storage, NSX-T overlay, and strict bank change windows.

Option A — Greenfield VCF 9 Build with Parallel Workload Migration (*Recommended*)

Deploy a net-new VCF 9 instance (management domain + workload domains) alongside the existing estate. Migrate workloads in controlled batches using vMotion / HCX / cold migration during approved change windows, then decommission legacy vCenter islands once validated.

Activities

- Validate all hosts against VCF 9 / ESXi 9 HCL (CPU generation, NIC, HBA, firmware); identify any requiring hardware refresh
- Validate SAN (FC or iSCSI — TBC) for VCF 9 supplemental storage compatibility; confirm vSAN readiness of management domain hosts
- Design single VCF 9 instance: one management domain (consolidating 3 legacy management vCenters), plus Production, UAT, and DR workload domains across 3 sites
- Deploy VCF 9 management domain (minimum 4 HCL-compliant hosts) via SDDC Manager bring-up
- Commission workload domains (Production, UAT, DR) and configure SAN as supplemental storage
- Design and deploy NSX 4.x overlay within VCF 9; map existing NSX-T segments and security policies to new constructs
- Configure DR and HA across all 3 sites per VCF 9 stretched/federated design
- Execute phased VM migration: prioritise UAT workloads first, then Production in sequenced bank change windows; DR workloads migrated last
- Perform application-level validation (UAT sign-off) per wave before decommissioning source vCenter islands
- Deliver structured SDDC Manager knowledge transfer and admin training throughout deployment

Advantages

- Zero risk of destabilising the existing production estate during build — legacy environment remains fully operational until cutover
- Allows thorough testing of VCF 9 platform (including SDDC Manager LCM, NSX 4.x, and SAN supplemental storage) before any production workload is moved
- Migration can be wave-sequenced to fit bank change windows without time pressure on the platform build itself
- Clean, supportable end-state with no legacy configuration debt carried into VCF 9
- NSX-T to NSX 4.x mapping can be validated in UAT domain before production cutover
- Aligns with VCF 9 greenfield bring-up best practice

Disadvantages

- Requires temporary parallel infrastructure capacity (hosts must serve both old and new environments during transition)

- Longer elapsed calendar time due to change window constraints on migration waves
- Requires careful network cutover planning at each wave to avoid production impact
- SDDC Manager operational learning curve must be managed in parallel with live migration activity

Relative Estimated Effort: High — but risk-adjusted effort is the lowest of the three options. Estimated 60–80 consultant man-days across all phases (see Effort Estimate section).

Option B — In-Place Consolidation (Migrate vCenters into VCF 9 Management Domain Sequentially)

Rather than building a fully parallel environment, progressively absorb existing vCenter islands into the VCF 9 management domain and workload domains by re-purposing hosts in-place, site by site.

Activities

- Evacuate and repurpose hosts from one site at a time: vMotion VMs to remaining sites, rebuild hosts to VCF 9 spec, re-introduce as VCF 9 workload domain hosts
- Consolidate management vCenters into SDDC Manager management domain incrementally
- Reconnect SAN storage to new VCF 9 workload domains
- Migrate NSX-T segments progressively to NSX 4.x as each site is absorbed
- Repeat per site until full estate is consolidated

Advantages

- Reduces need for net-new hardware (hosts are re-used progressively)
- Potentially lower hardware cost if existing hosts are HCL-compliant

Disadvantages

- Significantly higher operational risk: management plane visibility can be lost during vCenter consolidation if sequencing is incorrect
- Requires VMs to be live-migrated between sites during host evacuation, increasing network and change window pressure
- Any HCL failure discovered mid-sequence (CPU generation, NIC/HBA) halts the entire workstream
- NSX-T segment migration is more complex when done incrementally across a live estate
- SAN re-zoning and re-presentation must be coordinated with each host evacuation wave
- Not aligned with VCF 9 bring-up best practice; SDDC Manager bring-up expects a clean set of commissioned hosts
- Higher risk for a customer with no SDDC Manager experience operating a live environment during transition

Relative Estimated Effort: Very High — complexity and risk remediation overhead make this the most expensive option in practice despite apparent hardware savings. Estimated 80–100+ consultant man-days.

Option C — Phased Workload Domain Build (UAT-First Pilot, then Production)

A variant of Option A where the VCF 9 management domain is built first, followed immediately by the UAT workload domain as a live pilot. Lessons learned from the UAT migration directly inform the Production and DR domain build

and migration approach.

Activities

- Deploy VCF 9 management domain and UAT workload domain first
- Migrate all UAT VMs (~estimated subset of 3,500–4,000 total) into UAT workload domain; validate platform, SAN supplemental storage, NSX 4.x, and SDDC Manager operations
- Incorporate lessons learned; adjust Production workload domain design if required
- Deploy Production workload domain; execute production VM migration in change-window-aligned waves
- Deploy DR workload domain; configure cross-site DR and HA
- Decommission legacy vCenter islands progressively

Advantages

- UAT domain acts as a low-risk proving ground for VCF 9 platform, SAN integration, and NSX 4.x before any production workload is touched
- Builds customer SDDC Manager operational confidence in a non-production context before production cutover
- Reduces production migration risk by validating tooling, runbooks, and change window sequencing in UAT first
- Structured and auditable — well-suited to a financial institution's change governance requirements

Disadvantages

- Elapsed timeline is longer than Option A (sequential domain build adds calendar time)
- Requires discipline to not compress the UAT pilot phase under schedule pressure
- Still requires parallel capacity during transition (same constraint as Option A)

Relative Estimated Effort: High — broadly similar to Option A with a modest additional overhead for the structured pilot phase. Estimated 65–85 consultant man-days.

Options Comparison Summary

Criterion	Option A — Greenfield Parallel	Option B — In-Place Sequential	Option C — UAT-First Pilot
Production Risk	Low	High	Low–Medium
Hardware Requirement	Parallel capacity needed	Re-uses existing hosts	Parallel capacity needed
Change Window Pressure	Medium (waves planned)	High (live evacuation)	Medium
SDDC Manager Complexity	Managed	High (live environment)	Managed (UAT first)
NSX-T Migration Risk	Controlled	High	Controlled
SAN Integration Risk	Managed in design phase	Discovered mid-migration	Validated in UAT first
Estimated Effort (man-days)	60–80	80–100+	65–85

Criterion	Option A — Greenfield Parallel	Option B — In-Place Sequential	Option C — UAT-First Pilot
Recommended	☐ Yes	☐ No	△ Alternative if UAT pilot required

Recommended Approach

Option A — Greenfield VCF 9 Build with Parallel Workload Migration is the recommended approach, with elements of Option C's UAT-first sequencing applied within the migration wave plan.

The rationale is as follows:

- A greenfield build is explicitly aligned with VCF 9 bring-up best practice and eliminates the risk of destabilising the existing production estate during platform construction.
- With ~3,500–4,000 VMs and bank change windows, a well-sequenced wave plan (UAT waves first, then Production, then DR) is achievable within a realistic project timeline without the extreme complexity of in-place host evacuation.
- The confirmed NSX-T baseline eliminates NSX-V migration risk, making the network cutover to NSX 4.x within VCF 9 a controlled design exercise rather than an emergency remediation.
- SAN supplemental storage design can be validated and tested against the management domain before any workload domain is commissioned, reducing storage integration risk.
- The parallel build period provides the ideal structured environment for SDDC Manager knowledge transfer — the customer's operations team can shadow deployment activities and build operational confidence before taking ownership.
- Hardware HCL validation (CPU generation, NIC/HBA) must be completed in Phase 1 before any build activity begins; if a hardware refresh is required, it is scoped and procured before it becomes a critical-path blocker.

Recommended Approach

The recommended approach is a **greenfield VCF 9 build with phased brownfield workload migration**, deploying a single VCF 9 instance across all three sites with one management domain and three discrete workload domains (Production, UAT, DR). This optimises for lowest risk by avoiding in-place upgrade complexity, and for shortest timeline by enabling parallel workstream execution across the management domain build, NSX design, and storage integration tracks.

Overall Recommended Strategy

Build new, migrate in — do not attempt in-place upgrade of existing vCenter islands.

Given the brownfield estate comprises 9 standalone vCenter 8.x instances across 3 sites, with mixed vxLAN/VLAN networking, external SAN storage, and no existing SDDC Manager footprint, an in-place upgrade path does not exist. The greenfield VCF 9 instance will be commissioned alongside the existing estate, workloads migrated in waves, and legacy infrastructure decommissioned progressively. This approach preserves production availability

throughout and aligns with bank change window constraints.

Management Domain — Single Consolidated Domain Across 3 Sites

Recommended: Deploy a single VCF 9 management domain, consolidating the 3 existing management vCenters (one per site) into a single SDDC Manager-managed construct. Do not attempt to preserve 3 independent management vCenters within VCF 9.

- Assign the minimum 4 management domain hosts to a single management cluster, validated against the VCF 9 / ESXi 9 HCL before any deployment activity begins.
- Designate one primary site as the management domain anchor (the site with the most stable connectivity and lowest latency to the other two sites).
- The remaining two sites will be served by the workload domain vCenters deployed by SDDC Manager — they do not require independent management vCenters post-consolidation.
- Sequencing is critical: the management domain must be fully operational and stable before any workload domain commissioning begins, to avoid losing management plane visibility during transition.
- Hardware HCL validation for the 4 management domain hosts must be completed in Phase 1 (Assessment). If any hosts are pre-Skylake generation, a hardware refresh must be scoped and procured before deployment begins — this is a hard dependency.

Rationale: VCF 9 enforces a single management domain per instance managed by SDDC Manager. Attempting to maintain 3 parallel management vCenters introduces operational fragmentation and defeats the lifecycle management benefits of VCF 9. Consolidation is the only architecturally sound path.

Workload Domains — Dedicated Domains per Function, Deployed per Site

Recommended: Commission three workload domains under the single VCF 9 instance:

Workload Domain	Source vCenters	Primary Site	Notes
Production	2 existing production vCenters	Site A (primary)	Stretched or replicated across sites per DR design
UAT	2 existing UAT vCenters	Site B	Isolated from Production; lower change window sensitivity
DR	2 existing DR vCenters	Site C	Configured for failover target; HA/DR design drives cluster sizing

- Each workload domain will be built greenfield by SDDC Manager and validated before workload migration begins into that domain.
- UAT workload domain should be commissioned and migrated first — it carries lower business risk and provides a live rehearsal of the migration runbook before Production migration begins.
- Production migration follows UAT sign-off, executed in sequenced waves aligned to bank change windows.

- DR workload domain is configured in parallel with Production, with DR replication/failover testing completed before Production migration is declared complete.
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NSX Design — Migrate NSX-T Overlays into VCF 9 NSX

Recommended: Leverage the confirmed NSX-T foundation as the basis for the VCF 9 NSX 4.x design. Do not attempt a lift-and-shift of existing NSX-T configuration — redesign NSX segments, Tier-0/Tier-1 topology, and security policies natively within VCF 9, then cut over workloads to new segments as part of the migration wave.

- Conduct a full NSX-T inventory export (segments, logical routers, DFW rules, NAT, load balancer configs) during Phase 1 as a design input.
- Map existing vxLAN segments and VLAN-backed port groups to the target VCF 9 NSX segment design before any migration wave begins.
- Network cutover for each VM wave must be planned and tested in UAT before Production execution — this is the highest-risk step in each migration window.
- Retain existing NSX-T infrastructure in parallel until all workloads are migrated and validated; do not decommission until post-UAT sign-off.

Rationale: NSX-T being confirmed eliminates the NSX-V end-of-life migration risk. However, the mixed vxLAN/VLAN environment still requires careful segment mapping to avoid network disruption during VM migration. A clean redesign within VCF 9 NSX is lower risk than attempting to import legacy configuration.

Storage — SAN as Supplemental Storage, vSAN for Management Domain

Recommended: Retain external SAN (FC or iSCSI — protocol to be confirmed) as the primary datastore for workload domain VMs, configured as supplemental storage within VCF 9. Deploy vSAN for the management domain cluster only, as required by VCF 9 architecture.

- Management domain hosts must be validated for vSAN readiness (drive types, cache/capacity tier, disk group design) during Phase 1 hardware assessment.
- Workload domain hosts must have SAN connectivity (HBA/NIC HCL) validated against VCF 9 requirements before workload domain commissioning.
- SAN supplemental storage configuration must be explicitly designed and documented — some VCF 9 features (e.g. vSAN-native data services) will not be available on SAN-backed workload domains; this must be agreed with the customer during design.
- If the customer wishes to evaluate vSAN for workload domains in future, this should be noted as a post-project consideration and not in scope for this engagement.

Rationale: Forcing a full vSAN adoption across 60-70 workload hosts would require drive qualification, potential hardware refresh, and data migration from SAN — significantly extending the timeline and cost. Retaining SAN as supplemental storage is the lowest-risk path that preserves existing storage investments while achieving VCF 9 compliance.

VM Migration — Phased Wave Approach Aligned to Change Windows

Recommended: Execute VM migration in structured waves using vMotion (live migration) as the primary tool, with cold migration as a fallback for non-vMotion-eligible VMs. HCX should be evaluated during Phase 1 as an accelerator if cross-cluster L2 extension is required for network continuity during cutover.

- Wave sequencing: UAT VMs → Non-critical Production VMs → Tier-2 Production VMs → Tier-1 Production VMs.
 - Each wave must have a defined rollback procedure executable within the same change window.
 - At 3,500–4,000 VMs total, migration is achievable within a reasonable project timeline with proper wave planning — this is not the scale risk previously flagged and should not drive timeline pessimism.
 - A migration runbook must be produced and rehearsed in UAT before any Production wave executes.
 - Application-level validation (per the agreed acceptance criteria) must be completed for each wave before the next wave is approved to proceed.
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DR and HA — Three-Site Design

Recommended: Design DR and HA natively within VCF 9 using vSphere HA (within each workload domain cluster) for local HA, and a VMware-native DR mechanism (Site Recovery Manager or equivalent, to be confirmed during design) for cross-site DR between the Production and DR workload domains.

- HA must be enabled and validated on all workload domain clusters before Production migration begins.
 - DR replication configuration and failover testing must be completed and signed off before the Production migration is declared complete.
 - The three-site topology must be reflected in the NSX design (Tier-0 uplinks, BGP peering) to ensure network continuity during a site failover.
 - RPO and RTO targets must be agreed with the customer during Phase 1 and used to drive replication interval and cluster sizing decisions.
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Knowledge Transfer and Operational Readiness

Recommended: Embed structured SDDC Manager knowledge transfer throughout the engagement, not as a single end-of-project event.

- Phase 1: Deliver an SDDC Manager orientation session covering architecture, LCM concepts, and workload domain operations.
- Phase 2: Shadow-and-assist model during deployment — customer administrators participate in all SDDC Manager commissioning activities alongside the delivery team.
- Phase 3: Formal knowledge transfer sessions covering day-2 operations (LCM upgrades, host commissioning, workload domain expansion, certificate management, NSX day-2).
- Post-go-live: Include a defined hypercare period (recommended minimum 4 weeks) with escalation support to address operational queries as the team gains confidence.

Rationale: The customer has no prior SDDC Manager experience. Without structured, embedded knowledge transfer, post-handover operational risk is high — particularly for LCM operations where misconfiguration can affect the entire VCF instance. Embedding KT throughout delivery (rather than at the end) ensures the team is operationally ready at go-live.

Assumptions

The following assumptions underpin this Statement of Work. Any material deviation from these assumptions may result in a change request, revised timeline, or additional cost.

Platform & Architecture

- A single VCF 9 instance will be deployed comprising one management domain and separate workload domains for Production, UAT, and DR — not a multi-instance fleet.
- The VCF 9 environment will be built as a **greenfield deployment**; existing workloads will be migrated into VCF 9 workload domains rather than upgraded in-place from a prior VCF version.
- All 3 sites are on-premises data centres. No public cloud extension, hybrid cloud connectivity, or VMware Cloud on AWS is in scope at this stage.
- The 3 vCenter instances designated for the management domain represent one per site; VCF 9 will consolidate these into a single SDDC Manager-managed management domain construct.
- The 2 production, 2 UAT, and 2 DR vCenter instances will map to their respective VCF 9 workload domains as confirmed by the customer.
- VCF 9 licensing has been procured by the customer and is available prior to deployment commencement. Licensing activation and entitlement management are the customer's responsibility.

Software Versions

- All 9 existing vCenter instances are running vCenter 8.x and will remain at their current version throughout the assessment phase; no interim vCenter upgrades are in scope prior to VCF 9 migration.
- The existing vxLAN overlay is confirmed as **NSX-T** (not legacy NSX-V). NSX-V migration is explicitly excluded from scope.
- VCF 9 component versions (SDDC Manager, NSX 4.x, vSAN, ESXi 9.x) are assumed to be generally available, stable, and fully supported at the time of deployment. If a component version is not yet GA or carries a known critical defect, a project hold may be required.

Hardware

- All hosts (~64–74 total) are assumed to be on-premises across the 3 sites. Remote or co-location hosting arrangements are not in scope.
- Existing hosts are assumed to meet the VCF 9 / ESXi 9.x HCL requirements (CPU generation Skylake or later, supported NIC and HBA models, current firmware). **If hardware HCL validation during Phase 1 identifies non-compliant hosts, a hardware refresh will be required and is not included in this SOW — a change request will be raised.**
- The 4 management domain hosts are assumed to be HCL-compliant and available for VCF 9 management domain deployment. A minimum of 4 hosts per management domain cluster is required.
- Hardware refresh procurement, racking, cabling, and physical installation are the customer's responsibility and are excluded from this SOW.

Storage

- The customer's existing **external SAN** (Fibre Channel or iSCSI — protocol TBC) will be retained and configured as supplemental storage within VCF 9 workload domains. vSAN is not assumed to be the primary storage tier for workload domains.
- The SAN protocol (FC or iSCSI) will be confirmed by the customer prior to the start of Phase 1. This confirmation is required to complete VCF 9 storage design and HCL validation.
- vSAN readiness of hosts (drive types, cache/capacity tier availability, disk group design) will be assessed during Phase 1. If vSAN is required for the management domain, hosts must carry qualifying NVMe or SSD drives; any remediation is the customer's responsibility.
- SAN zoning, fabric configuration, LUN provisioning, and storage array management remain the customer's responsibility throughout the engagement.

Networking

- The customer will provide and maintain all physical network infrastructure (switches, routers, firewalls, load balancers) required to support VCF 9 deployment, including required VLANs, MTU settings (≥ 9000 for vSAN/overlay), and uplink bandwidth.
- Firewall rule changes required to support VCF 9 management plane, NSX overlay, and inter-site replication traffic will be implemented by the customer's network/security team in a timely manner. Delays in firewall provisioning are a project risk owned by the customer.
- Active Directory (AD) / LDAP identity sources for vCenter and SDDC Manager SSO integration will be provided and maintained by the customer. AD schema changes or new service accounts required will be provisioned by the customer prior to deployment.
- DNS and NTP services are available, reliable, and correctly configured across all 3 sites prior to VCF 9 deployment commencing.

VM Migration

- The confirmed VM count is **3,500–4,000 VMs** across the estate. The previously cited figure of 35,000–40,000 VMs is retracted and does not form the basis of this SOW.
- VM migrations will be executed using vMotion and/or VMware HCX (tooling to be confirmed during Phase 1). Cold migration may be used for non-critical workloads where live migration is not feasible.
- Migration sequencing will be planned around the customer's **bank change windows**. The customer will provide a confirmed change window schedule prior to Phase 2 commencement. Narrow or infrequent windows will extend the migration timeline and are a customer-owned constraint.
- **Application-level validation is the customer's responsibility.** The customer's application and business owners will perform functional testing to confirm applications are operating as normal or better post-migration. The consultant team will provide infrastructure-level validation only.
- The customer will identify and provide application owners, migration wave approvers, and a change advisory board (CAB) contact for each migration wave.

DR & HA

- DR and HA design will span all 3 sites as a confirmed requirement. Specific RPO/RTO targets will be agreed and documented during Phase 1 (Assessment & Design). This SOW assumes RPO/RTO targets are achievable within the constraints of the existing inter-site network bandwidth; if bandwidth is insufficient, network upgrades are the customer's responsibility.
- Inter-site replication and failover tooling (e.g. vSphere Replication, SRM, or VCF-native capabilities) will be confirmed during Phase 1. Licensing for any additional DR tooling beyond VCF 9 entitlements is the customer's responsibility.

Operations & Knowledge Transfer

- The customer has **no prior SDDC Manager experience**. Structured knowledge transfer, administrator training, and/or extended hypercare covering SDDC Manager LCM, workload domain operations, and day-2 procedures are included in this SOW scope and are a mandatory deliverable.
- The customer will nominate a minimum of 2 internal administrators to participate in knowledge transfer sessions and shadow deployment activities throughout the engagement.
- Post-handover operational support (BAU support, break-fix, ongoing LCM upgrades beyond the scope of this engagement) is the customer's responsibility. Extended managed services or support retainers are not included unless separately contracted.

General

- Active VMware support and warranty entitlements are in place for all existing vCenter, ESXi, and NSX-T components for the duration of the engagement.
- The customer will provide timely access to all required environments, systems, and documentation. Delays caused by the customer's inability to provide access may result in timeline and cost impacts.
- All work will be performed remotely unless otherwise agreed. On-site travel, if required, will be subject to a separate expenses agreement.
- This SOW does not include any application re-platforming, containerisation, Tanzu/VKS deployment, Aria Automation, or Horizon EUC workstreams unless explicitly added via change request.

Dependencies

Customer Dependencies

- **Hardware inventory and HCL validation data** must be provided by ACME prior to the Assessment phase commencing. This includes CPU generation, NIC model, HBA model, and firmware versions for all ~64-74 hosts across the three sites. Failure to provide this promptly will delay design finalisation and may surface a mandatory hardware refresh requirement.
- **SAN protocol confirmation (FC or iSCSI)** must be provided by ACME before storage design can be completed. The protocol directly affects VCF 9 HCL validation, host adapter requirements, and the supplemental storage design approach.

- **Network topology documentation** — full current-state diagrams and configuration exports for the existing NSX-T overlay and VLAN environment across all three sites must be supplied by ACME before NSX 4.x design work begins.
- **vCenter and NSX-T export data** — configuration exports (clusters, port groups, distributed switches, NSX-T segments, routing, firewall rules) from all 9 vCenter instances must be made available to the delivery team during the Assessment phase.
- **Change Advisory Board (CAB) / change window schedule** must be shared with the project team at engagement kick-off. Migration sequencing and the overall project timeline will be planned around ACME's bank change windows, and any changes to the schedule must be communicated with sufficient lead time.
- **Named technical contacts** — ACME must designate a technical lead and a network/storage SME who are available throughout the engagement to provide environment access, answer design queries, and participate in design review workshops.
- **Application owner availability** — application owners or representatives must be available during migration waves and UAT to perform application-level validation, as success is defined by applications functioning as normal or better post-migration.
- **SDDC Manager training participants** must be identified and confirmed prior to the Knowledge Transfer phase. ACME must allocate sufficient time for designated administrators to attend structured training and shadowing sessions.
- **VCF 9 licensing confirmation** — ACME must confirm that VCF 9 licences are procured, allocated, and available for activation prior to the Deployment phase commencing.
- **Hardware refresh procurement** (if required following HCL assessment) must be initiated and completed by ACME before affected hosts can be onboarded into VCF 9 clusters. Delays in procurement will directly impact the deployment timeline.
- **Access and credentials** — ACME must provide the delivery team with appropriate administrative access to all vCenter instances, NSX-T managers, physical hosts, SAN management interfaces, and network infrastructure required to perform assessment and deployment activities.
- **Site access and data centre coordination** — physical access to all three sites must be arranged by ACME for any activities requiring hands-on hardware work (e.g., host commissioning, cabling validation, firmware updates).

Third-Party Dependencies

- **SAN/storage vendor** — if SAN firmware or driver updates are required to meet the VCF 9 HCL, the storage vendor must be engaged by ACME to provide supported firmware, drivers, and any required professional services. This is outside the scope of this engagement.
- **Server hardware vendor** — if a hardware refresh is required for pre-Skylake hosts, ACME must engage the hardware vendor for procurement, delivery, and rack/stack. Lead times for server hardware must be factored into the project schedule.
- **Network infrastructure vendor** — any physical switching or routing changes required to support VCF 9 underlay design (e.g., MTU configuration, VLAN provisioning, BGP/OSPF changes) are the responsibility of ACME's network team or their incumbent network vendor.

- **VMware/Broadcom support** — given VCF 9 is a recent release, ACME must maintain an active Broadcom support contract to enable timely resolution of any product defects or early-release issues encountered during deployment. Escalation to Broadcom GSS may be required and is outside the control of the delivery team.
- **Change management / CAB approval** — all deployment and migration activities are subject to ACME's internal change management process. Delays in CAB approval are outside the delivery team's control and must be managed by ACME's project manager.

Risks & Challenges

Technical Risks & Challenges

The following risks have been identified based on the confirmed estate details and engagement scope. Each risk has been assessed for impact and, where applicable, a mitigation approach is noted. These risks must be reviewed and formally accepted or mitigated during the Assessment & Design phase.

Hardware & Infrastructure Risks

- **Hardware HCL Compatibility (CPU Generation)** — ESXi 9 drops support for pre-Skylake CPU generations. If any of the ~64-74 hosts across the estate are on older hardware, a mandatory hardware refresh is required before VCF 9 deployment can begin. This represents a potential major cost and timeline impact if discovered late in the engagement. HCL validation must be the first activity in Phase 1.
 - **Hardware HCL Compatibility (NIC / HBA)** — VCF 9 imposes strict NIC and HBA HCL requirements. Existing network and storage adapters must be individually validated. Non-compliant adapters may require card-level replacement even where the host chassis is otherwise acceptable.
 - **vSAN Readiness of Management Domain Hosts** — VCF 9 management domain requires vSAN-ready hosts (appropriate NVMe/SSD drive types, cache and capacity tier configuration, disk group design). The current estate uses external SAN; it is unknown whether the 4 management domain hosts carry qualifying local drives. This must be assessed and resolved before management domain deployment.
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Storage Architecture Risks

- **SAN as Supplemental Storage in VCF 9** — VCF 9 workload domains are architecturally optimised for vSAN. Retaining the existing SAN (FC or iSCSI — protocol TBC) requires explicit design as supplemental storage and may restrict access to certain VCF 9 features. The SAN protocol, array model, and multipathing configuration must be confirmed and HCL-validated. Failure to agree this in the design phase could result in rework during deployment.
 - **SAN Protocol Confirmation Outstanding** — The SAN protocol (Fibre Channel or iSCSI) has not yet been confirmed. This directly affects VCF 9 storage design, HBA HCL requirements, and network fabric design. This is a blocking dependency for storage design sign-off.
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Network & NSX Risks

- **NSX Migration Complexity from Mixed vxLAN / VLAN Environment** — The existing estate uses a mixed overlay (NSX-T confirmed) and VLAN environment. All existing NSX-T logical constructs (segments, Tier-0/Tier-1 gateways, distributed firewall rules, load balancer configurations) must be fully mapped and re-expressed in the VCF 9 NSX 4.x design. Incomplete mapping is a leading cause of network outages during VM migration cutover.
 - **Network Cutover Risk During VM Migration** — Moving VMs from standalone vCenter islands into VCF workload domains requires coordinated network cutover. Any mismatch between source and target segment configuration could result in production connectivity loss. Detailed per-application network dependency mapping and a tested rollback procedure are mandatory for each migration wave.
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Management Domain Consolidation Risks

- **Consolidation of 3 Management vCenters into a Single VCF Management Domain** — VCF 9 operates a single management domain construct governed by SDDC Manager. Mapping three existing site-level management vCenters (one per site assumed) into this single construct requires careful sequencing to avoid loss of management plane visibility during transition. This is a non-trivial workstream and must be treated as a critical-path activity with a tested rollback plan at each step.
 - **VCF 9 Release Maturity** — VCF 9 is a recent release. Component versions (SDDC Manager, NSX 4.x, vSAN, ESXi 9) must be confirmed as generally available, fully patched, and stable prior to deployment. Early-release defects could affect delivery confidence and may require patch-level adjustments mid-engagement. A pre-deployment release review and VMware GSS engagement plan should be established.
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Migration Execution Risks

- **VM Migration Scale (3,500–4,000 VMs)** — While the corrected VM count (3,500–4,000) is materially more manageable than earlier estimates, it remains a significant migration scope. Sequencing across Production, UAT, and DR workload domains, combined with application dependency mapping and per-wave validation, requires disciplined wave planning and tooling (HCX and/or vMotion).
 - **Bank Change Window Constraints** — As a financial institution, ACME operates under strict, infrequent change windows (typically weekend-only). Migration velocity is directly constrained by the number and duration of approved windows. Wave planning must be designed around this constraint from the outset, and the project timeline must reflect realistic window availability. Failure to account for this will result in schedule overrun.
 - **Workload Domain Migration Complexity** — Migrating VMs from 9 standalone vCenter instances into structured VCF workload domains (Production, UAT, DR) requires per-VM or per-application assessment of compute, storage, and network dependencies. Applications with hard storage affinity to SAN LUNs or with complex NSX micro-segmentation policies require individual migration runbooks.
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DR & HA Risks

- **DR and HA Design Across 3 Sites** — Delivering DR and HA across all three sites within a single VCF 9 instance adds significant design complexity. Stretched cluster viability, inter-site latency, and witness placement for vSAN (if used in management domain) must all be validated. RPO/RTO targets must be formally agreed and documented before DR architecture is finalised, as they directly determine the replication and failover technology choices.
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Operational Readiness Risks

- **No SDDC Manager Experience — Post-Handover Operational Risk** — ACME has no prior experience operating SDDC Manager. VCF 9 day-2 operations — including lifecycle management (LCM), workload domain commissioning, host remediation, and certificate management — are all SDDC Manager-driven. Without structured knowledge transfer and hands-on training included in this engagement, the risk of misconfigured LCM operations or failed post-handover upgrades is high. Structured KT and an extended hypercare period are included in scope to mitigate this risk.
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Risk Summary Table

#	Risk	Likelihood	Impact	Phase Where Risk Materialises
1	Hardware HCL — CPU generation incompatibility	Medium	High	Assessment
2	Hardware HCL — NIC/HBA incompatibility	Medium	Medium	Assessment
3	vSAN readiness of management domain hosts	Medium	High	Assessment / Design
4	SAN supplemental storage design complexity	Medium	Medium	Design
5	SAN protocol unconfirmed (blocking dependency)	High	Medium	Assessment
6	NSX-T to NSX 4.x construct remapping	Medium	High	Design / Migration
7	Network cutover risk during VM migration	Medium	High	Migration
8	3-site management domain consolidation	Medium	High	Deployment
9	VCF 9 release maturity / early-release defects	Low-Medium	Medium	Deployment
10	VM migration scale (3,500–4,000 VMs)	Low	Medium	Migration
11	Bank change window constraints on migration velocity	High	High	Migration
12	3-site DR/HA design complexity	Medium	High	Design
13	No SDDC Manager operational experience	High	High	Post-Handover

Note: Risks rated High impact must have a documented mitigation plan agreed with ACME before the relevant phase commences. Hardware HCL validation (Risks 1–3) is a hard dependency for all subsequent phases and must be completed within the first two weeks of Phase 1.

Exclusions

The following items are explicitly out of scope for this engagement unless otherwise agreed in writing via a signed Change Request:

Platform & Infrastructure

- In-place upgrade of any existing vCenter 8.x instance to VCF 9; the engagement covers greenfield VCF 9 build and workload migration only
- Upgrade or remediation of any third-party hardware not on the VCF 9 / ESXi 9 HCL; hardware refresh procurement, racking, and cabling is a customer responsibility
- Physical network infrastructure design, reconfiguration, or implementation (e.g. top-of-rack switch configuration, inter-site WAN/MPLS changes, firewall rule implementation)
- SAN array firmware upgrades, SAN fabric zoning changes, or LUN provisioning beyond what is required to present storage to VCF 9 hosts; SAN infrastructure management remains with the customer
- vSAN design, deployment, or migration — this SOW assumes SAN (FC or iSCSI) is retained as the primary/supplemental storage model for workload domains
- Design or deployment of any public cloud extension, hybrid cloud connectivity, or VMware Cloud on AWS / Azure VMware Solution integration

Networking & Security

- Firewall policy migration, rule translation, or implementation on any physical or virtual firewall platform (e.g. Palo Alto, Fortinet, Cisco ASA)
- NSX-V to NSX-T migration — NSX-T is confirmed in use; NSX-V remediation is not in scope
- Redesign of the customer's physical underlay network or IP addressing scheme beyond what is required for VCF 9 overlay integration
- Active Directory (AD) redesign, domain consolidation, or DNS/DHCP infrastructure changes beyond VCF 9 integration requirements
- PKI infrastructure build or certificate authority (CA) redesign; the SOW covers VCF 9 certificate integration with an existing CA only

Workload & Application

- Application-level remediation, re-platforming, or code changes required as a result of the migration
- Operating system upgrades or patching of guest VMs (Windows, Linux, etc.) before, during, or after migration
- Database migration, schema changes, or application reconfiguration post-migration
- End-user support or service desk augmentation during the migration period
- Validation of third-party application functionality beyond the agreed acceptance criteria (applications functioning as normal or better post-migration); deep application testing remains the customer's responsibility
- Decommissioning or physical removal of legacy infrastructure after migration is complete

Disaster Recovery & HA

- Procurement, licensing, or deployment of any third-party DR tooling (e.g. Zerto, Veeam, Commvault) not included in the VCF 9 bundle
- DR runbook authoring, DR automation scripting, or orchestration beyond what is delivered as part of the VCF 9 DR workload domain design
- Live DR failover testing against production workloads during the engagement; DR testing is limited to the agreed UAT/DR validation activities defined in the project plan

Organisational & Operational

- Formal VMware-accredited training courses or certification preparation; structured knowledge transfer and admin enablement are in scope, but vendor-delivered training is a separate commercial arrangement
- ITSM/CMDB updates, change record management, or configuration item (CI) population resulting from the migration
- Post-handover BAU operational support, break-fix, or managed services beyond the agreed hypercare period
- Change Advisory Board (CAB) submission preparation or change record ownership; the customer retains responsibility for all change management processes
- Staffing or backfill of customer operational teams during the engagement period

Deliverables

Phase 1 — Assessment & Design

- **Hardware Compatibility Assessment Report** — documented validation of all ~64–74 hosts against the VCF 9 / ESXi 9 HCL, covering CPU generation (Skylake or later), NIC models, HBA models, and firmware levels; includes a clear pass/fail matrix and hardware refresh recommendations where required
- **vSAN Readiness Report** — assessment of drive types, cache/capacity tier suitability, and disk group design options for management domain hosts; includes recommendation on vSAN vs. SAN supplemental storage model for workload domains
- **SAN Storage Design Decision Record** — documents the agreed storage architecture (SAN as supplemental storage or primary storage within VCF 9 workload domains), SAN protocol confirmation (FC or iSCSI), and any VCF 9 feature limitations that apply
- **NSX-T Current-State Assessment** — inventory and mapping of existing NSX-T vxLAN segments and VLAN topology across all 3 sites; input to VCF 9 NSX 4.x target design
- **VCF 9 High-Level Design (HLD)** — covers single VCF instance architecture, management domain consolidation (3 existing management vCenters → single SDDC Manager-managed construct), workload domain topology (Production, UAT, DR), multi-site HA and DR design across 3 sites, NSX 4.x overlay/underlay design, and storage architecture
- **VCF 9 Low-Level Design (LLD)** — detailed configuration specifications for SDDC Manager, vCenter, NSX 4.x, vSAN (management domain), SAN supplemental storage (workload domains), host profiles, cluster sizing, and

network design per site

- **Architecture Diagrams** — logical and physical topology diagrams covering: multi-site VCF instance layout, workload domain structure, NSX 4.x overlay design, SAN/vSAN storage connectivity, and DR/HA topology across all 3 sites
 - **Risk Assessment & Assumptions Register** — consolidated register of all identified technical risks (HCL gaps, NSX migration complexity, SAN/vSAN design, management domain consolidation, change window constraints) with impact ratings, mitigations, and open assumptions requiring customer confirmation
 - **Migration Options Analysis** — documented evaluation of 2–3 migration approach options (see Migration Approach section) with advantages, disadvantages, effort comparison, and a clear recommended approach
 - **Migration Plan** — phased migration sequencing for 3,500–4,000 VMs across Production, UAT, and DR workload domains; aligned to bank change window constraints; includes tooling selection (HCX, vMotion, cold migration), wave planning, rollback procedures, and dependency mapping
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Phase 2 — Deployment & Migration

- **Management Domain Build Runbook** — step-by-step implementation runbook for greenfield VCF 9 management domain deployment across 3 sites, including SDDC Manager installation, vCenter commissioning, and NSX 4.x management plane configuration
 - **Workload Domain Build Runbooks** — individual runbooks for commissioning Production, UAT, and DR workload domains within SDDC Manager, including cluster configuration, storage attachment (SAN supplemental), and NSX segment creation
 - **NSX 4.x Migration Runbook** — detailed procedure for migrating existing NSX-T vxLAN segments and VLAN configurations into the VCF 9 NSX 4.x instance, including segment mapping, policy migration, and cutover steps
 - **VM Migration Wave Runbooks** — per-wave execution runbooks covering pre-migration checks, migration execution steps (HCX/vMotion/cold), network cutover procedures, post-migration validation steps, and rollback criteria; one runbook per agreed migration wave
 - **DR & HA Configuration Runbook** — implementation runbook for cross-site DR and HA configuration, including stretched cluster or site-recovery design (per agreed HLD), failover group definitions, and RPO/RTO configuration
 - **Change Request Documentation** — pre-formatted change request artefacts aligned to the customer's bank change management process, covering each migration wave and infrastructure change activity
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Phase 3 — UAT, Knowledge Transfer & Closeout

- **Test Plan & UAT Scripts** — structured test cases covering: management domain operations, workload domain provisioning, VM connectivity post-migration, NSX 4.x overlay functionality, SAN storage access, DR failover testing, and HA event simulation across all 3 sites
- **UAT Sign-Off Report** — formal record of UAT execution results against the agreed acceptance criteria (applications functioning as normal or better post-migration), including any outstanding defects and resolution status
- **DR Failover Test Report** — documented results of at least one planned DR failover test per site pair, including RPO/RTO measurements against agreed targets

- **As-Built Documentation** — final as-built configuration record for the VCF 9 environment, capturing actual deployed state of SDDC Manager, all workload domains, NSX 4.x, storage, and network configuration; deviations from LLD noted and explained
- **SDDC Manager Operations Guide** — customer-facing operational runbook covering day-2 operations: LCM (lifecycle management) patching and upgrade workflows, workload domain commissioning and decommissioning, host addition/removal, SDDC Manager backup and restore, and alert/event management
- **Knowledge Transfer Delivery Record** — structured training and knowledge transfer sessions covering SDDC Manager administration, VCF 9 LCM operations, NSX 4.x day-2 management, and DR/HA operational procedures; includes attendance records and session materials
- **Hypercare Support Summary** — documented record of issues raised and resolved during the post-handover hypercare period, with a formal handover sign-off to BAU support
- **Project Closure Report** — executive summary of delivered scope, open items, risks retired, outstanding recommendations, and formal project sign-off

Acceptance Criteria

Acceptance of each phase and the overall engagement is conditional on the customer's formal sign-off against the criteria defined below. All validation activities must be completed within agreed bank change windows and documented in a Validation & Acceptance Report before sign-off is requested.

AC-1 — Phase 1: Assessment & Design Acceptance

#	Criterion	Evidence Required
1.1	Hardware HCL Assessment Report delivered and reviewed	Signed-off HCL Compatibility Report covering all ~64-74 hosts (CPU generation, NIC, HBA, firmware)
1.2	SAN storage design decision confirmed (supplemental vSAN or SAN-principal)	Written customer confirmation of storage model; reflected in HLD
1.3	NSX-T overlay mapping completed and validated against VCF 9 NSX design	Network Design document accepted by customer network team
1.4	VCF 9 High-Level Design (HLD) accepted	Customer-signed HLD covering management domain, workload domains (Prod, UAT, DR), multi-site topology, and DR/HA design
1.5	Low-Level Design (LLD) accepted	Customer-signed LLD including SDDC Manager configuration, vSAN/SAN storage design, NSX segment mapping, and workload domain specifications
1.6	Migration Runbook v1.0 accepted	Runbook reviewed and approved by customer change management team
1.7	Risk Register reviewed and accepted	All open risks acknowledged and mitigations agreed in writing

AC-2 — Phase 2: VCF 9 Management Domain Acceptance

#	Criterion	Evidence Required
2.1	SDDC Manager deployed and accessible across all 3 sites	Screenshot/screen-recording of SDDC Manager UI; health dashboard showing green status
2.2	Management domain vCenter operational	vCenter login confirmed; all management hosts visible and in connected state
2.3	NSX Manager deployed and management overlay segments operational	NSX Manager UI accessible; control-plane health confirmed
2.4	Management domain HA validated	Simulated host failure test: management VMs restart within agreed RTO; test result documented
2.5	SDDC Manager LCM baseline established	Bundle repository configured; baseline inventory scan completed with no critical alerts
2.6	SAN supplemental storage (if applicable) presented and accessible to management domain	Datastore visible in vCenter; test VM provisioned and powered on

AC-3 — Phase 2: Workload Domain Acceptance (Production, UAT, DR)

#	Criterion	Evidence Required
3.1	All workload domains (Prod, UAT, DR) commissioned in SDDC Manager	SDDC Manager inventory showing all domains in healthy state
3.2	Workload domain vCenters operational and clusters populated	vCenter login; host count matches design; no disconnected hosts
3.3	NSX segments and distributed firewall policies migrated and operational	Connectivity test matrix completed (East-West and North-South); no unplanned policy drops
3.4	SAN datastores presented and accessible per workload domain	Datastore mount confirmed; IOPS/throughput baseline test passed
3.5	DR workload domain reachable from all 3 sites	Cross-site ping/connectivity matrix documented

AC-4 — Phase 2: VM Migration Acceptance

Given the corrected scope of 3,500–4,000 VMs migrated across bank change windows, acceptance of each migration wave is required before the subsequent wave commences.

#	Criterion	Evidence Required
4.1	Pre-migration baseline captured for each wave	Application health check report (ping, service status, application URL response) signed off before migration window opens

#	Criterion	Evidence Required
4.2	VMs successfully registered and powered on in target workload domain	vCenter inventory screenshot; VM power state = On; no VMware Tools disconnected alerts
4.3	Application-level validation passed	Application owner sign-off per wave: application functions as normal or better post-migration (login, transaction, API response, batch job)
4.4	Network connectivity confirmed post-migration	Connectivity test matrix: VM-to-VM, VM-to-physical, VM-to-external; no unplanned packet loss
4.5	No P1/P2 incidents attributable to migration open at wave close	Incident tracker reviewed; any open incidents resolved or formally accepted as pre-existing
4.6	Rollback not invoked	If rollback was invoked, root cause documented and remediated before wave is re-attempted

Note: Application-level validation is the responsibility of the customer's application owners. Broadcom/VMware Professional Services will provide the migration runbook and coordinate the validation window; application sign-off must be provided by the customer within the agreed window.

AC-5 — DR and HA Validation

#	Criterion	Evidence Required
5.1	HA failover test — intra-site	Simulated host failure; VMs restart on surviving hosts within agreed RTO; test report signed off
5.2	Cross-site DR failover test	Planned DR test: workloads failed over to DR site; applications validated at DR site by application owners; RTO/RPO targets met and documented
5.3	Cross-site DR failback test	Workloads failed back to primary site; application validation repeated; no data loss beyond agreed RPO
5.4	DR runbook accepted	Customer DR team confirms runbook is accurate and executable without PS assistance

Assumption: Specific RPO and RTO targets have not yet been confirmed. These must be agreed during Phase 1 (Design) and documented in the LLD before DR validation criteria are finalised.

AC-6 — Knowledge Transfer and Operational Readiness Acceptance

Given the customer has no prior SDDC Manager experience, operational readiness acceptance is a mandatory gate before project closure.

#	Criterion	Evidence Required
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6.1	SDDC Manager administration training delivered	Attendance register; training completion acknowledgement signed by attendees
6.2	LCM (patching/upgrade) procedure demonstrated and practised	Customer admin successfully executes a non-production LCM update under PS supervision; outcome documented
6.3	Day-2 operations runbook delivered and accepted	Customer signs off Day-2 Operations Guide covering: workload domain management, host commissioning/decommissioning, NSX operations, SAN storage management, and DR procedures
6.4	Hypercare period completed	All hypercare support tickets resolved or formally transitioned to BAU support; hypercare closure report signed
6.5	As-Built Documentation accepted	Customer signs off As-Built document reflecting final deployed configuration

AC-7 — Overall Project Acceptance

Final project sign-off requires **all** of the following:

- All phase acceptance criteria (AC-1 through AC-6) formally signed off by the designated customer project sponsor
- No open Severity 1 or Severity 2 defects attributable to the VCF 9 build or migration
- As-Built Documentation, DR Runbook, and Day-2 Operations Guide delivered and accepted
- Validation & Acceptance Report compiled and signed by both parties

Sign-off must be provided in writing (email or formal sign-off document) by the customer's nominated Technical Authority and Project Sponsor within **5 business days** of each acceptance gate being presented. Failure to respond within this window, with no documented objection, will be treated as acceptance by default.

Estimated Professional Services Effort

Activity	Effort (man-days)
Phase 1 - Assessment & Discovery	
Hardware HCL validation (CPU generation, NIC/HBA, firmware) across ~64-74 hosts	5-7
vSAN readiness assessment & SAN storage design review (FC/iSCSI TBC)	3-4
NSX-T overlay & VLAN inventory mapping (existing vxLAN segments, routing, security policies)	4-5
Network topology assessment (3-site, inter-site connectivity, MTU, VLAN tagging)	2-3
VM inventory & workload classification (3,500-4,000 VMs across 9 vCenters)	3-4

Activity	Effort (man-days)
Current-state management domain analysis (3 management vCenters, consolidation planning)	3-4
DR & HA requirements capture (RPO/RTO targets, site topology, failover design inputs)	2-3
Change window & migration sequencing constraints workshop	1-2
Assessment report & risk register	2-3
Phase 1 Total	25-35
Phase 2 - Architecture & Design	
VCF 9 logical design (single instance, management domain + Prod/UAT/DR workload domains)	5-7
Management domain consolidation design (3 sites → single SDDC Manager construct)	4-5
NSX 4.x design (overlay segments, T0/T1 topology, security policy migration from NSX-T)	5-7
SAN supplemental storage design for VCF 9 workload domains	3-4
vSAN management domain storage design & disk group specification	2-3
3-site DR & HA design (stretched clusters and/or cross-site failover topology)	5-7
Workload domain design (Prod, UAT, DR — host allocation, cluster sizing, resource pools)	4-5
Migration approach design & wave planning (vMotion/HCX sequencing, change window alignment)	3-4
HLD & LLD documentation	5-7
Design review & sign-off workshop	1-2
Phase 2 Total	37-51
Phase 3 - Deployment & Build	
VCF 9 management domain deployment (SDDC Manager, vCenter, NSX, vSAN — 4 hosts)	5-7
Production workload domain build & commissioning (2 vCenters → VCF WLD, ~30-35 hosts)	7-10
UAT workload domain build & commissioning (2 vCenters → VCF WLD)	5-7
DR workload domain build & commissioning (2 vCenters → VCF WLD, 3-site integration)	5-7
NSX 4.x deployment & policy migration (overlay, micro-segmentation, distributed firewall)	7-10
SAN supplemental storage integration & datastore provisioning per workload domain	4-5
3-site DR & HA configuration (cross-site networking, failover policies, replication)	5-7
SDDC Manager LCM configuration & operational runbook setup	3-4
Phase 3 Total	41-57
Phase 4 - Workload Migration	

Activity	Effort (man-days)
Migration tooling setup & validation (HCX and/or vMotion, wave sequencing)	3-4
Wave 1 migration — UAT VMs (validation wave, lower risk)	5-7
Wave 2 migration — Production VMs (change-window-constrained, sequenced batches)	10-15
Wave 3 migration — DR / remaining VMs	5-7
Post-migration network cutover & validation per wave	4-5
Application-level validation support (per acceptance criteria)	3-5
Phase 4 Total	30-43
Phase 5 - UAT, Knowledge Transfer & Hypercare	
Structured SDDC Manager admin training (LCM, workload domain ops, day-2 operations)	3-5
Operational runbooks & knowledge transfer documentation	3-4
Hands-on KT sessions (SDDC Manager, NSX 4.x, vSAN, DR failover procedures)	3-4
Failover test execution & documentation (DR validation across 3 sites)	3-4
UAT support & defect resolution	3-5
Extended hypercare (post-go-live, remote support)	5-10
Phase 5 Total	20-32
Project Management	
Project management, stakeholder reporting, risk & issue management (full engagement duration)	15-20
Project Management Total	15-20
TOTAL ESTIMATED EFFORT	168-238 man-days

Notes:

- Effort ranges reflect uncertainty in hardware HCL outcomes, SAN protocol confirmation (FC vs iSCSI), and the number of change windows available for migration waves. If hardware refresh is required, Phase 3 effort will increase and should be re-scoped separately.
- Migration effort (Phase 4) assumes 3,500-4,000 VMs sequenced across bank change windows using HCX and/or vMotion. If change window frequency is lower than assumed, elapsed calendar time will extend without materially increasing man-day effort.
- DR & HA design and build effort (Phases 2-3) will be refined once RPO/RTO targets and inter-site connectivity specifications are confirmed.
- Extended hypercare duration (Phase 5) is indicative at 4-6 weeks post-handover; this should be agreed during contract negotiation given the customer's zero SDDC Manager experience.

- This estimate covers professional services delivery effort only and excludes hardware procurement, software licensing, travel & expenses, and any third-party vendor engagement.

Indicative Costing

Important Disclaimer

All figures below are **indicative only** and are provided for budgetary planning purposes. Final pricing will be confirmed following completion of the Assessment & Design phase, hardware HCL validation, SAN protocol confirmation, and NSX-T topology discovery. Rates are exclusive of VAT, travel and expenses, and any third-party tooling or licensing costs.

Indicative Rate Card

Resource Role	Indicative Day Rate (GBP)	Notes
Principal Architect (VCF / NSX)	£1,800 – £2,200	Design authority, management domain consolidation, NSX-T overlay design
Senior Infrastructure Engineer	£1,400 – £1,700	VCF deployment, workload domain build, SAN integration
NSX-T Network Engineer	£1,500 – £1,800	NSX-T migration mapping, segment design, DFW policy migration
Migration Engineer	£1,200 – £1,500	VM migration execution, HCX configuration, sequencing
DR / SRM Engineer	£1,400 – £1,700	Multi-site DR design, replication configuration, failover testing
Training / KT Consultant	£1,200 – £1,500	SDDC Manager training, runbook authoring, knowledge transfer sessions
Project Manager	£1,200 – £1,500	Delivery governance, change window scheduling, RAID management

Blended team day rate assumed at approximately **£1,500/day** for scenario-level estimates below.

Effort Summary by Phase (Man-Days)

Phase	Activity	Estimated Man-Days
Phase 1 – Assessment & Design	HCL validation (64–74 hosts), SAN protocol/HCL review, NSX-T topology discovery, vCenter island mapping (9 instances), management domain consolidation design, workload domain design (Prod/UAT/DR), multi-site DR/HA design, storage design (SAN supplemental model)	35 – 50

Phase	Activity	Estimated Man-Days
Phase 2 – Greenfield VCF 9 Build	Management domain deployment (4 hosts, 3-site aware), SDDC Manager configuration, NSX-T integration, SAN supplemental storage configuration, workload domain commissioning (Prod, UAT, DR), multi-site HA/DR configuration	40 – 55
Phase 2 – VM Migration	Migration tooling setup (HCX/vMotion), wave planning and sequencing (3,500–4,000 VMs), network cutover coordination, bank change window execution across multiple waves	30 – 45
Phase 3 – UAT, DR Testing & KT	Application validation support, DR failover testing (3-site), SDDC Manager admin training, runbook and operational documentation, hypercare period	20 – 30
Total Estimated Effort		125 – 180 man-days

Note: The upper bound reflects scenarios where hardware refresh is required, SAN integration is complex (FC zoning), or NSX-T segment count is high. The lower bound assumes HCL-compliant hardware, straightforward iSCSI SAN, and a well-structured NSX-T environment.

Indicative Fixed-Price Scenarios

Scenario	Description	Indicative Cost (GBP)	Key Variables
Scenario A – Optimistic	HCL-compliant hardware (no refresh), iSCSI SAN, clean NSX-T topology, ~125 man-days	£185,000 – £200,000	All hosts pass HCL; SAN integration straightforward; NSX-T segment count manageable
Scenario B – Base Case	Partial HCL remediation, FC SAN, moderate NSX-T complexity, ~155 man-days	£225,000 – £260,000	Some hosts require NIC/HBA firmware remediation; FC zoning effort; moderate DFW policy migration
Scenario C – Complex	Hardware refresh required for subset of hosts, complex FC SAN, high NSX-T segment/DFW complexity, extended migration waves due to change window constraints, ~180 man-days	£265,000 – £310,000	Pre-Skylake hosts identified; complex FC fabric; large NSX-T segment/policy estate; migration extends across many change windows

Excluded from all scenarios: Hardware procurement and refresh costs, VMware/Broadcom licensing (customer has purchased VCF 9 licensing), third-party tooling licences (e.g. HCX Enterprise if not included in VCF 9 bundle), travel and subsistence, and any application remediation required post-migration.

Cost Confidence and Next Steps

- Cost confidence at this stage is assessed as **Medium** (±25%). Confidence will increase to **High** (±10%) following:
- Completion of the **Hardware HCL Assessment** (CPU generation, NIC/HBA models confirmed for all 64-74 hosts)
 - Confirmation of **SAN protocol** (FC vs iSCSI) and array HCL status
 - **NSX-T topology discovery** (segment count, DFW rule count, routing topology)
 - Confirmation of **management domain multi-site topology** (stretched vs federated)
 - Agreement on **migration wave structure** relative to available bank change windows

A revised fixed-price proposal will be issued following Phase 1 (Assessment & Design) sign-off.

Project Timeline

The following timeline is indicative and based on confirmed project facts, assumptions, and risk factors. Final durations will be refined during Phase 1 (Assessment & Design) once hardware HCL validation, NSX-T topology mapping, SAN protocol confirmation, and workload domain sizing are complete. All change activity is subject to ACME's bank change window constraints.

Note: Durations below assume no major hardware refresh is required. If pre-Skylake hosts are identified during HCL validation, Phase 1 will be extended and Phase 2 start will be deferred accordingly. A hardware refresh workstream would be tracked as a parallel dependency.

Phase Overview

Phase	Description	Indicative Duration	Key Dependency
Phase 1	Assessment, HCL Validation & Design	8-10 weeks	Hardware inventory, SAN protocol confirmation, NSX-T topology export
Phase 2	Greenfield VCF 9 Build — Management Domain	4-6 weeks	Phase 1 design sign-off, HCL-compliant hardware available
Phase 3	Workload Domain Deployment (Prod, UAT, DR)	6-8 weeks	Management domain stable, SAN supplemental storage configured
Phase 4	Workload Migration (3,500-4,000 VMs)	10-16 weeks	Workload domains live, HCX or vMotion tooling deployed, change windows confirmed
Phase 5	DR / HA Validation Across 3 Sites	3-4 weeks	All workload domains populated, inter-site networking validated
Phase 6	UAT, Knowledge Transfer & Hypercare	4-6 weeks	Migration complete, SDDC Manager training delivered
	Total Indicative Duration	35-50 weeks	Subject to hardware refresh outcome and change window cadence

Phase 1 — Assessment, HCL Validation & Design (Weeks 1-10)

- Conduct full hardware inventory across all ~64–74 hosts; validate against VCF 9 / ESXi 9 HCL (CPU generation, NIC, HBA, firmware)
 - Confirm SAN protocol (FC or iSCSI) and validate SAN HCL for VCF 9 supplemental storage integration
 - Export and document existing NSX-T topology (segments, T0/T1 gateways, distributed firewall rules) across all 9 vCenter islands
 - Map existing mixed vxLAN/VLAN networking to target VCF 9 NSX-T design
 - Produce VCF 9 Architecture Design Document (ADD) covering: single instance, management domain, Production/UAT/DR workload domains, 3-site HA/DR topology
 - Define management domain consolidation approach (3 existing management vCenters → single SDDC Manager construct)
 - Agree storage design: vSAN for management domain minimum; SAN as supplemental storage for workload domains
 - Identify and agree VM migration wave plan (sequencing across change windows)
 - **Gate:** Design sign-off by ACME and architecture review before Phase 2 commences
-

Phase 2 — Greenfield VCF 9 Management Domain Build (Weeks 11-16)

- Deploy SDDC Manager and management domain (minimum 4 HCL-validated hosts)
 - Configure management domain vCenter, NSX-T Manager cluster, and vSAN (management domain)
 - Validate SDDC Manager LCM, commissioning workflows, and day-2 operational tooling
 - Establish inter-site management plane connectivity across all 3 sites
 - **Gate:** Management domain health check and SDDC Manager operational validation
-

Phase 3 — Workload Domain Deployment (Weeks 17-24)

- Commission Production workload domain (2 existing vCenter islands → 1 VCF workload domain)
 - Commission UAT workload domain (2 existing vCenter islands → 1 VCF workload domain)
 - Commission DR workload domain (2 existing vCenter islands → 1 VCF workload domain)
 - Integrate SAN supplemental storage into each workload domain
 - Deploy and validate NSX-T overlay networking per workload domain
 - Configure cross-site HA and DR networking foundations
 - **Gate:** All workload domains healthy and storage/networking validated before migration commences
-

Phase 4 — Workload Migration (Weeks 25-40)

- Execute VM migration waves per agreed sequencing plan (3,500–4,000 VMs across Production, UAT, DR domains)
- Migration tooling: HCX and/or vMotion (cold migration for non-HCX-eligible workloads)
- Each wave subject to ACME bank change window scheduling — weekend windows assumed
- Application-level validation performed post-wave before proceeding to next wave

- Parallel decommission of source vCenter islands as waves complete
 - **Note:** Migration duration is the most variable phase; 10-16 weeks assumes consistent change window availability and no major application remediation
-

Phase 5 — DR / HA Validation Across 3 Sites (Weeks 38-44)

- Execute DR failover and failback tests across all 3 sites
 - Validate HA behaviour under simulated host and site failure scenarios
 - Document RPO/RTO results against agreed targets
 - Remediate any gaps identified during testing
 - **Gate:** Signed DR test report accepted by ACME
-

Phase 6 — UAT, Knowledge Transfer & Hypercare (Weeks 41-50)

- Structured SDDC Manager admin training delivered to ACME operations team (covering LCM, workload domain commissioning, day-2 operations)
 - Hands-on knowledge transfer sessions: NSX-T management in VCF 9, SAN supplemental storage operations, SDDC Manager upgrade workflows
 - Customer UAT: application functionality validation against acceptance criteria
 - Hypercare period: consultant available for operational queries and issue resolution post-handover
 - Final documentation handover: as-built, runbooks, operational procedures
 - **Gate:** ACME sign-off on UAT completion and operational readiness
-

Key Timeline Risks & Assumptions

Risk / Assumption	Timeline Impact
Pre-Skylake hosts identified requiring hardware refresh	+4-8 weeks (Phase 1 extended; Phase 2 deferred)
SAN protocol or HCL incompatibility discovered	+2-4 weeks design rework
Bank change windows limited to fortnightly weekend slots	Migration phase extends toward 16-week upper bound
NSX-T topology complexity requiring significant redesign	+2-3 weeks Phase 1
VCF 9 early-release stability issues	Potential hold on Phase 2/3 pending patch availability
SDDC Manager training not completed before handover	Hypercare period extended at additional cost

Assumption: Timeline assumes ACME provides hardware inventory data, SAN protocol confirmation, and NSX-T topology exports within the first 2 weeks of Phase 1. Delays in information provision will shift all subsequent phase start dates.